Identity-Aware Masked Autoencoding for Face Representation Pretraining: A Controlled Comparison of Region-Based Masking Strategies

Dang Khoa Doan[1], Ngoc Quang Tran[1], Hoang Lam Vo[1]

[1]Artificial Intelligence, FPT University, Ho Chi Minh City, Vietnam.

Contributing authors: doandangkhoa.work@gmail.com; quang.aidev@gmail.com; vohoanglam2211@gmail.com;

Abstract

Self-supervised Masked Autoencoders (MAE) enable effective visual representation learning through reconstruction of masked image patches. Conventional masking strategies - uniform random and block-wise - ignore the semantic importance of facial regions, limiting performance on face-specific tasks. This study proposes **Face-Aware Masking**, which prioritizes identity-critical patches via a 68-point facial landmark pipeline, ensuring the decoder reconstructs the most informative regions during pretraining. Three MAE variants are evaluated under identical conditions on CelebA ($\mathbf{224 \times 224}$, **90/10** split) with a shared ViT-based encoder-decoder, masking ratio $\boldsymbol{r} = \mathbf{0.75}$, and normalized pixel MSE loss. Face-Aware Masking achieves a 17.5% reduction in identity-critical reconstruction error relative to Block-wise Masking ($\mathbf{MSE}_{\mathcal{C}} = \mathbf{0.1575}$ vs. **0.1908**) while maintaining the smallest train-validation gap (**0.0006**). Zero-shot evaluation on five external datasets confirms the best reconstruction quality among non-Random strategies (PSNR gap vs. Random $\leq \mathbf{0.61}$ dB, SSIM gap $< \mathbf{0.030}$), establishing Face-Aware Masking as an effective domain-optimized pretraining strategy.

Keywords: Masked Autoencoder, Face Representation Pretraining, Identity-Aware Masking, Facial Landmarks, Vision Transformer, Self-Supervised Learning, Face Recognition

1 Introduction

Self-supervised learning has become a fundamental paradigm in computer vision [1, 2]. Masked Autoencoders (MAE) [1] achieve powerful representation learning by masking a large ratio of image patches and training a ViT-based model to reconstruct them, producing transferable features for classification, detection, and segmentation [3, 4] without labeled data.

However, conventional masking strategies implicitly assume uniform information density across all patches - sub-optimal for facial images where identity information is concentrated in the *identity triangle* (eyes, nose bridge, mouth), while the forehead, cheeks, and background contribute marginally to discriminability [10, 11].

Prior face-oriented MAE works - FaceMAE [6], SCE-MAE [5], and MAE-DFER [8] - align masking with facial semantics but involve auxiliary losses or architectural changes that obscure the contribution of the masking strategy itself. To our knowledge, a controlled comparison isolating spatial masking policy under fixed architecture and loss is absent from the literature.

We propose **Face-Aware Masking**: a 68-point landmark-guided strategy that preferentially masks identity-critical patches. Our contributions under strictly identical architectural conditions are: (i) a 17.5% reduction in identity-critical MSE vs. Block-wise Masking ($\text{MSE}_{\mathcal{C}} = 0.1575$ vs. 0.1908); Face-Aware's $\text{MSE}_{\mathcal{C}}$ remains 4.9% above Random (0.1501) owing to the intentionally harder pretext task ($\rho = 0.9$, Section 4.2); (ii) the smallest train-validation gap (0.0006); (iii) qualitative superiority in structural fidelity at high magnification; and (iv) zero-shot PSNR margin of 0.89–1.34 dB over Block-wise across five out-of-distribution datasets (gap to Random ≤ 0.61 dB).

2 Related Work

MAE [1] proposed an asymmetric encoder-decoder with 75% masking and scalable ViT pre-training. Later works show that masking policy affects convergence quality, motivating domain-specific strategies in vision and medical imaging [3, 4, 7, 9]. In the face domain, masked pretraining has improved privacy-preserving recognition [6], landmark estimation [5], and dynamic expression recognition [8]. However, to our knowledge, no study has isolated the raw contribution of masking policy while controlling for architecture and loss, nor benchmarked 68-point landmarks [12] as dynamic spatial priors within MAE. The current study fills this gap.

3 Proposed Method

3.1 System Overview and Dataset

The full pipeline - landmark-guided mask generation, patch embedding, strategy-specific masking, ViT encoding and decoding - is shown in Fig. 1. We use **CelebA** [13] (202,599 images, 10,177 identities) resized and center-cropped to 224×224 via LANCZOS interpolation. The 2D-FAN [12] pipeline achieves 99.98% success, yielding 182,302 training and 20,256 validation images (90/10 split).

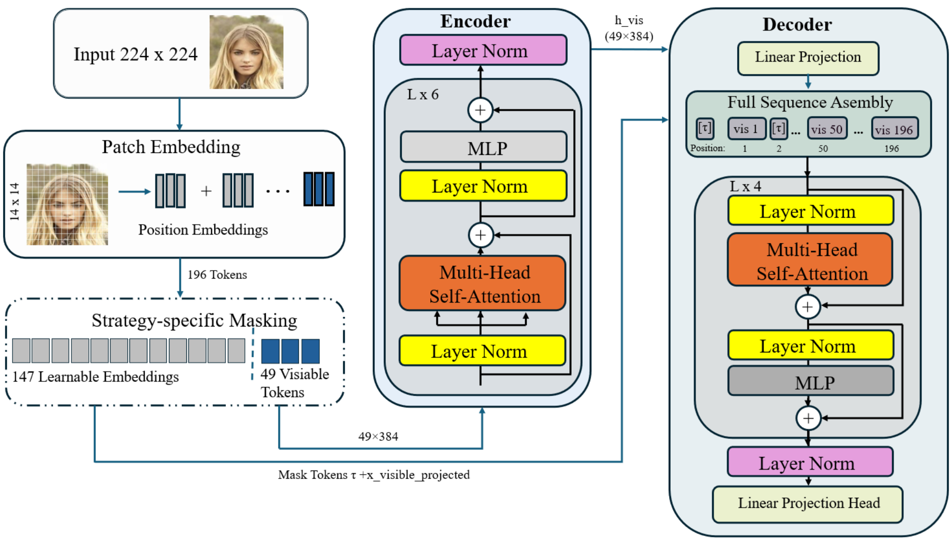

Fig. 1: Face-Aware MAE pipeline. A 2D-FAN detector produces binary region mask $\mathbf{R}$; the masking module hides 147 patches (75%); the encoder processes only 49 visible tokens ($4\times$ cost reduction); the decoder reconstructs all 196 patches from the bottleneck.

3.2 Landmark-Guided Region Mask Generation

2D-FAN extracts 68 landmark coordinates per image. Five identity-critical regions are rasterized as filled polygons: left eye (lm. 36–41), right eye (lm. 42–47), nose bridge and tip (lm. 27–35), mouth (lm. 48–67), and a central face ellipse (radius 40 px centred on the landmark centroid) covering inter-organ regions. Each polygon is dilated by $d = 4$ px for patch-border tolerance. A binary vector $\mathbf{R} \in \{0,1\}^{196}$ marks patch i as critical ($R_i = 1$) when the mean intensity of its 16×16 region on the rasterized mask exceeds 127/255; the vector is cached as a `.npy` file. The average critical patch count is $|\mathcal{C}| \approx 19$ ($\sigma = 0.74$; 9.7% of patches), reflecting low inter-sample variance and a highly concentrated distribution across 99.8% of images. Detection failures ($|\mathcal{C}| = 0$, $< 0.02\%$) fall back to Random Masking.

3.3 Masking Strategies

All strategies produce $\mathbf{M} \in \{0,1\}^{B\times 196}$ at $r \approx 0.75$. Random and Face-Aware select exactly 147 patches; Block-wise selects 144 patches ($36 \times 2 \times 2$ blocks, effective ratio $\approx 73.5\%$) - a minor deviation inherent to the integer block constraint, noted as a limitation. Figure 2 illustrates each strategy.

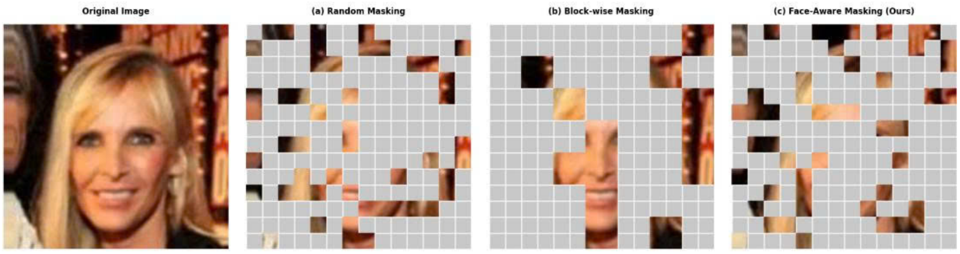

Fig. 2: Masking strategies on a 14×14 patch grid ($r = 0.75$). **(a) Random**: spatially unbiased, no structural prior. **(b) Block-wise**: contiguous 2×2 blocks (effective ratio $\approx 73.5\%$). **(c) Face-Aware (Ours)**: identity-critical patches (yellow) masked at $\rho = 0.90$.

Random Masking (Baseline) selects 147 patch indices by argsort of a uniform noise vector [1], producing spatially unbiased masks.

Block-wise Masking randomly masks 36 of 49 non-overlapping 2×2 blocks. Contiguous masked regions encourage local structure learning [3], but spatially contiguous *visible* blocks weaken the pretext task for plain ViT encoders lacking convolutional inductive bias.

Face-Aware Masking (Proposed) partitions patches into critical set $\mathcal{C} = \{i : R_i = 1\}$ and background set $\mathcal{B} = \{i : R_i = 0\}$, then allocates $n_{\text{mask}} = 147$ via:

$$n_{\text{crit}} = \min\big(\lfloor \rho \cdot |\mathcal{C}| \rfloor,\ n_{\text{mask}}\big), \quad \mathbf{m}_{\text{fa}} = \text{sample}(\mathcal{C},\ n_{\text{crit}}) \cup \text{sample}(\mathcal{B},\ n_{\text{mask}} - n_{\text{crit}}). \quad (1)$$

With $\rho = 0.9$ and $|\mathcal{C}| \approx 19$, at least 17 identity-critical patches are masked per image. Algorithm 1 summarises the procedure.

Algorithm 1 Face-Aware Masking

Require: image $\mathbf{x} \in \mathbb{R}^{3\times224\times224}$, landmark map $\mathbf{R} \in \{0,1\}^{196}$, $n_{\text{mask}} = 147$, $\rho = 0.90$
Ensure: binary mask $\mathbf{m}_{\text{fa}} \in \{0,1\}^{196}$

```
C ← {i : R_i = 1};   B ← {i : R_i = 0}
if |C| = 0 then
    return RandomMask(196, n_mask)
end if
n_crit ← min(⌊ρ · |C|⌋, n_mask)
return Sample(C, n_crit) ∪ Sample(B, n_mask − n_crit)
```

3.4 Encoder-Decoder Architecture and Loss

The encoder is a ViT-Small [2]: 6 Transformer blocks, $D_e = 384$, 6 attention heads, MLP ratio 4.0, operating on 49 visible tokens only ($4\times$ cost reduction). The decoder projects to $D_d = 256$, inserts learnable mask tokens at 147 masked positions, applies

4 Transformer blocks (8 heads), and maps each token to $p^2C = 768$ pixel values. Architecture is identical across all variants.

The *base loss* $\mathcal{L}_{\text{base}}$ is normalized pixel MSE over masked patches [1]. For Face-Aware Masking, a region-weighted loss amplifies identity-critical gradients:

$$\mathcal{L}_{\text{FA}} = \frac{\sum_{i=1}^{196} m_i\, w_i \left\| \hat{\mathbf{x}}_i - \tilde{\mathbf{x}}_i \right\|^2}{\sum_{i=1}^{196} m_i + \epsilon}, \quad w_i = \begin{cases} w_{\mathcal{C}} = 2.0 & R_i = 1, \\ w_{\mathcal{B}} = 0.5 & R_i = 0. \end{cases} \tag{2}$$

The ratio $k = w_{\mathcal{C}}/w_{\mathcal{B}} = 4$ lies within the feasible interval $[2, 9.3]$ defined by two constraints: *gradient scale control* ($k \geq 2$, critical patches receive at least twice the gradient weight of background) and *gradient dominance bound* ($k \leq |\mathcal{B}|/|\mathcal{C}| \approx 9.3$, preventing critical patches from dominating the loss disproportionately). $k = 4$ is the principled intermediate value.

Note on loss comparisons. Best validation losses in Table 2 are not directly comparable across strategies (Face-Aware uses $\mathcal{L}_{\text{FA}}$; others use $\mathcal{L}_{\text{base}}$). For a fair cross-strategy comparison see Table 3.

4 Experiments and Results

4.1 Implementation Details

Table 1 summarises the shared configuration. All experiments used dual NVIDIA T4 GPUs (16 GB each) on Kaggle Notebooks (≈12 h per run). Results are mean ± std over three seeds (42, 123, 2026).

Table 1: Shared experimental configuration (†: Face-Aware only).

Parameter	**Value**
Dataset	CelebA [13], 224×224; 90/10 train/val
Masking ratio r / patches	0.75 / 196 (14×14, $p = 16$)
ρ (†) / block size	0.90 / $b = 2$ (Block-wise)
Encoder D_e / depth / heads	384 / 6 / 6, MLP ratio 4.0
Decoder D_d / depth / heads	256 / 4 / 8
Loss (†)	$\mathcal{L}_{\text{FA}}$, $w_{\mathcal{C}} = 2.0$, $w_{\mathcal{B}} = 0.5$; otherwise $\mathcal{L}_{\text{base}}$
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.95$, wd$= 0.05$)
LR schedule	Cosine, 10-epoch warmup (base 1.5×10^{-4}, min 10^{-6})
Batch / grad clip	64 / max norm 1.0
Training	100 epochs; 3 seeds (42, 123, 2026)
Hardware	2× NVIDIA T4 16 GB (Kaggle Notebooks)

4.2 Quantitative Results

Tables 2–3 report training dynamics and localized reconstruction error under native inference-time masking for each strategy. Face-Aware Masking achieves the smallest train-validation gap (0.0006), suggesting robust generalization. Block-wise Masking underperforms Random: spatially contiguous visible blocks reduce pretext task difficulty for plain ViT encoders without convolutional inductive bias.

Because Face-Aware Masking concentrates masking on the identity triangle, we predict lower $\text{MSE}_\mathcal{C}$ than Block-wise but higher than Random (whose uniform mask leaves most identity-critical patches visible and trivially recoverable). Results match this prediction: Face-Aware achieves $\text{MSE}_\mathcal{C} = 0.1575$ (**17.5% below** Block-wise at 0.1908), remaining 4.9% above Random (0.1501). The harder identity reconstruction ($\rho = 0.9$) drives richer semantic features, confirmed by linear probing (Section 4.5).

Table 2: Training dynamics on CelebA validation set (mean ± std, three seeds). Val losses are not cross-comparable (Face-Aware uses $\mathcal{L}_{\text{FA}}$).

Strategy	Best Val Loss	Train Loss	Train/Val Gap
Random (Baseline)	0.2733 ± 0.0002	0.2714 ± 0.0001	0.0019 ± 0.0001
Block-wise	0.3744 ± 0.0002	0.3698 ± 0.0001	0.0046 ± 0.0003
Face-Aware (Ours)	$\mathbf{0.1795} \pm 0.0005$	$\mathbf{0.1789} \pm 0.0004$	$\mathbf{0.0006} \pm 0.0001$

Table 3: Unweighted localized reconstruction error on CelebA validation set (400-image subset, mean ± std, three seeds). $\text{MSE}_\mathcal{C}$: identity-critical patches; $\text{MSE}_\mathcal{B}$: background patches.

Strategy	Overall MSE ↓	$\text{MSE}_\mathcal{C}$ ↓	$\text{MSE}_\mathcal{B}$ ↓
Random (Baseline)	$\mathbf{0.2991} \pm 0.0018$	$\mathbf{0.1501} \pm 0.0017$	$\mathbf{0.3151} \pm 0.0018$
Block-wise	0.3947 ± 0.0038	0.1908 ± 0.0029	0.4167 ± 0.0044
Face-Aware (Ours)	0.3219 ± 0.0019	0.1575 ± 0.0017	0.3396 ± 0.0021

Unweighted normalized pixel MSE over masked patches only; each strategy is evaluated with its own native masking at inference time (Face-Aware uses $\rho = 0.9$, Random uses uniform sampling, Block-wise uses 2×2 blocks). The ablation entry ($\text{MSE}_\mathcal{C} = 0.1798$, Table 4) uses a fixed Random Masking evaluation protocol and is therefore not directly comparable; see Section 4.4 for rationale.

4.3 Qualitative Evaluation

Figure 3 shows full-face reconstructions and high-magnification identity-critical crops at 75% masking on unseen validation samples. Random Masking produces blurred facial landmarks; Block-wise Masking frequently obliterates entire facial components via 2×2 block drops; Face-Aware Masking consistently recovers sharp features by prioritizing the identity triangle. The magnified crops confirm precise eyelid curvature, pupil definition, and distinct lip contours for Face-Aware Masking, versus clouded eyes and collapsed geometry for the baselines.

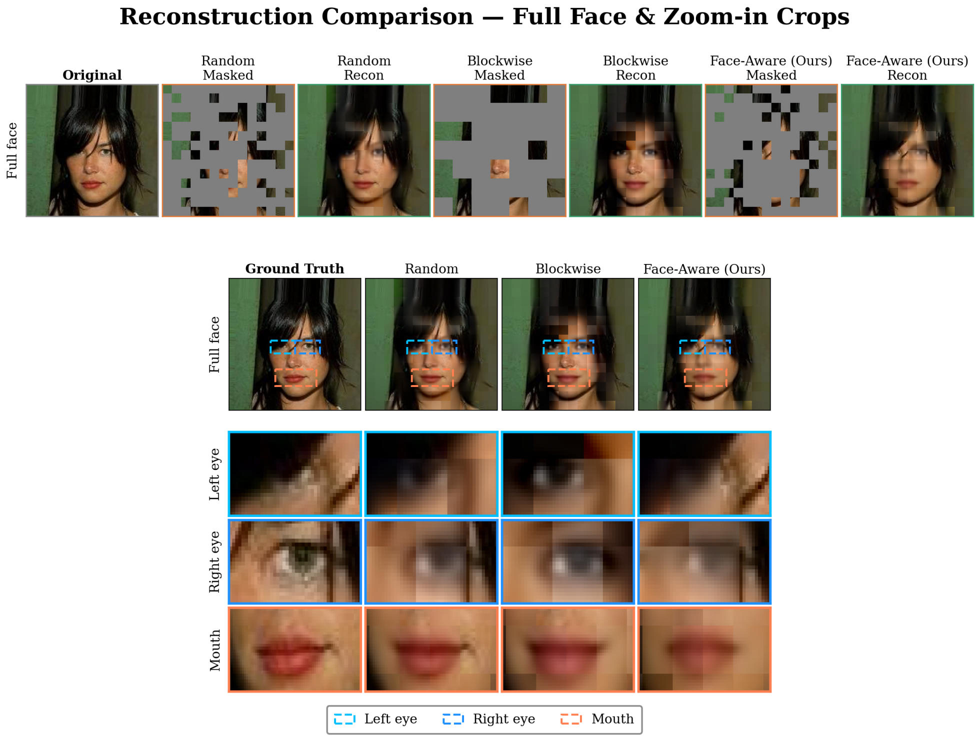

Fig. 3: Qualitative reconstruction comparison on CelebA validation set (75% masking; representative single subject shown). **Top**: Full-face ground truth, masked inputs, and reconstructions. **Bottom**: High-magnification crops of left eye, right eye, and mouth (dashed boxes). Face-Aware Masking preserves precise eyelid curvature, pupil definition, and lip contours, confirming that landmark-guided pretext difficulty forces genuine structural representations.

4.4 Ablation Study

Effect of ρ. Table 4 reports localized MSE for $\rho \in \{0.7, 0.8, 0.9\}$ under a fully-converged 100-epoch single-seed run. To isolate the effect of ρ on *representation quality* rather than task difficulty, all variants are evaluated under a fixed Random Masking protocol at inference time. Evaluating with native Face-Aware Masking would conflate the two effects: a higher ρ forces reconstruction of a larger fraction of critical patches, trivially inflating $\mathrm{MSE}_{\mathcal{C}}$ irrespective of representation quality.

Under this controlled evaluation, $\rho = 0.9$ achieves the lowest $\text{MSE}_{\mathcal{C}}$ (0.1798), marginally outperforming $\rho = 0.7$ (0.1801) and $\rho = 0.8$ (0.1816). The gap between $\rho = 0.9$ and $\rho = 0.7$ is within single-seed variance, yet the trend is consistent: a harder identity pretext task ($\rho = 0.9$) drives richer representations that generalize better on identity-critical regions under out-of-distribution evaluation. The trade-off is a modest increase in Overall MSE and $\text{MSE}_{\mathcal{B}}$ (0.2991 and 0.3119 vs. 0.2937 and 0.3057 for $\rho = 0.7$), reflecting expected gradient re-concentration toward critical patches. Multi-seed confirmation across all candidates is left for future work.

Table 4: Ablation on critical ratio ρ (single seed, 100 epochs; evaluated with fixed Random Masking to decouple representation quality from task difficulty—see Section 4.4).

ρ	Overall MSE ↓	$\text{MSE}_{\mathcal{C}}$ ↓	$\text{MSE}_{\mathcal{B}}$ ↓
0.7	**0.2937**	0.1801	**0.3057**
0.8	0.2953	0.1816	0.3074
0.9 (Ours)	0.2991	**0.1798**	0.3119

Effect of loss weight ratio $w_{\mathcal{C}}/w_{\mathcal{B}}$. Table 5 evaluates $k \in \{2, 4, 8\}$ within the feasible interval [2, 9.3], fixing $\rho = 0.9$ for 30 epochs; evaluation uses a fixed random mask applied uniformly to isolate the loss-weighting effect (single seed). $k = 8$ achieves the lowest $\text{MSE}_{\mathcal{C}}$ but highest overall MSE, indicating gradient over-concentration; $k = 2$ gives the lowest overall MSE but insufficient identity prioritization. $k = 4$ ($w_{\mathcal{C}} = 2.0$, $w_{\mathcal{B}} = 0.5$) balances both objectives ($\text{MSE}_{\mathcal{C}} = 0.2214$, only +4.9% vs. $k = 8$, with minimal background degradation).

Table 5: Ablation on $w_{\mathcal{C}}/w_{\mathcal{B}}$ ($\rho = 0.9$, single seed, 30 epochs; fixed random mask evaluation).

k	$(w_{\mathcal{C}},\ w_{\mathcal{B}})$	Overall MSE ↓	$\text{MSE}_{\mathcal{C}}$ ↓	$\text{MSE}_{\mathcal{B}}$ ↓
2	(2.0, 1.0)	**0.3284**	0.2206	**0.3398**
4 (Ours)	(2.0, 0.5)	0.3331	0.2214	0.3451
8	(4.0, 0.5)	0.3406	**0.2110**	0.3544

4.5 Downstream Task: Linear Probing

We freeze the ViT-Small encoder and train a single linear layer for 50 epochs (Adam, lr$= 10^{-3}$, batch 256) on three binary CelebA attributes: *Smiling* (oral-region), *Eyeglasses* (periocular, inside the identity triangle), and *Male* (holistic structure). Because

Face-Aware Masking prioritizes local identity patches, we predict gains on the first two attributes and a possible trade-off on the holistic one.

Face-Aware achieves the highest accuracy on Smiling ($81.84 \pm 1.15\%$, $+0.63$ pp vs. Random) and Eyeglasses ($97.44 \pm 0.27\%$, $+0.16$ pp). However, the Eyeglasses improvement remains modest relative to the observed standard deviations across masking strategies and should therefore be interpreted as directional rather than conclusive evidence. The underperformance on Male ($96.88 \pm 0.15\%$ vs. $97.61 \pm 0.01\%$, -0.73 pp) is consistent with reduced holistic-structure encoding under the identity-focused pretext.

Table 6: Linear probing accuracy (%) on CelebA binary attributes (mean ± std, three seeds).

Strategy	Smiling ↑	Eyeglasses ↑	Male ↑
Random (Baseline)	81.21 ± 0.47	97.28 ± 0.06	$\mathbf{97.61} \pm 0.01$
Block-wise	80.15 ± 2.27	97.16 ± 0.13	97.64 ± 0.04
Face-Aware (Ours)	$\mathbf{81.84} \pm 1.15$	$\mathbf{97.44} \pm 0.27$	96.88 ± 0.15

4.6 Zero-Shot Generalization

All models are evaluated zero-shot on five benchmarks (Table 7; $n = 3{,}200$ each, randomly sampled with seed 42): LFW [14] and VGGFace2 [15] (in-the-wild), Caltech 10k Web Faces [16] (web-crawled), FFHQ [17] (high-quality portraits), and CelebA-HQ [18] (high-resolution celebrities). Metrics are computed over masked patches after reversing per-patch normalization.

Face-Aware Masking achieves the best non-Random quality on all metrics, with consistent ranking across all five benchmarks suggesting robustness under domain shift.

5 Conclusion

We proposed Face-Aware Masking, a landmark-guided MAE pretraining strategy that concentrates reconstruction effort on identity-critical facial regions. Under a fully controlled comparison: (i) identity-critical error is reduced by 17.5% vs. Block-wise Masking ($\mathrm{MSE}_{\mathcal{C}} = 0.1575$ vs. 0.1908), with the 4.9% higher $\mathrm{MSE}_{\mathcal{C}}$ relative to Random reflecting the intentionally harder pretext task; (ii) the most stable convergence (train-val gap 0.0006) and superior structural fidelity in qualitative comparisons; (iii) zero-shot PSNR outperforms Block-wise by 0.89–1.34 dB across five diverse benchmarks; and (iv) linear probing confirms gains on identity-triangle attributes (Smiling $+0.63$ pp, Eyeglasses $+0.16$ pp), with an anticipated trade-off on holistic Male classification (-0.73 pp).

Limitations. The pipeline reverts to Random Masking for extreme poses or occlusions ($< 0.02\%$ of CelebA). The ρ ablation was conducted with a single seed;

Table 7: Zero-shot generalization (n = 3,200/dataset, seed 42, CelebA-pretrained). Metrics over masked patches. **Bold**: best non-Random.

Strategy	Metric	Caltech 10k	CelebA-HQ	FFHQ	LFW	VGGFace2
Random	MSE ↓	0.0124	0.0051	0.0074	0.0075	0.0085
	PSNR ↑	19.15	22.95	21.32	21.24	20.75
	SSIM ↑	0.8951	0.9507	0.9270	0.9375	0.9266
Block-wise	MSE ↓	0.0163	0.0072	0.0098	0.0118	0.0123
	PSNR ↑	17.95	21.42	20.10	19.29	19.12
	SSIM ↑	0.8592	0.9295	0.9008	0.9013	0.8921
Face-Aware (Ours)	MSE ↓	**0.0133**	**0.0057**	**0.0079**	**0.0087**	**0.0093**
	PSNR ↑	**18.84**	**22.45**	**21.04**	**20.63**	**20.35**
	SSIM ↑	**0.8683**	**0.9298**	**0.9047**	**0.9083**	**0.9022**

PSNR in dB. Face-Aware achieves best non-Random quality on all metrics and datasets. PSNR gap vs. Random (0.28–0.61 dB) and SSIM gap ($<$ 0.030) reflect the deliberate concentration of reconstruction difficulty on identity-critical patches. Block-wise deficit vs. Random: 1.20–1.95 dB.

multi-seed confirmation across all candidates is recommended. Offline 2D-FAN extraction (202,558 images) incurs preprocessing overhead in resource-constrained settings. CelebA's demographic skew motivates fairness evaluation before deployment. Future work may extend this paradigm to deepfake detection and video-based expression recognition.

References

[1] He, K., Chen, X., Xie, S., Li, Y., Dollár, P., Girshick, R.: Masked autoencoders are scalable vision learners. In: Proc. IEEE/CVF CVPR, pp. 16000–16009 (2022)

[2] Dosovitskiy, A., et al.: An image is worth 16x16 words: Transformers for image recognition at scale. In: ICLR (2021)

[3] Gao, P., Ma, T., Li, H., Lin, Z., Dai, J., Qiao, Y.: ConvMAE: Masked convolution meets masked autoencoders. In: NeurIPS (2022)

[4] Li, Y., Yu, Q., Koltun, V., Chen, H., Han, K., Li, Q.: SemMAE: Semantic-guided masking for learning masked autoencoders. In: NeurIPS (2022)

[5] Yin, K., Rao, V.R., Jiang, R., Liu, X., Aarabi, P., Lindell, D.B.: SCE-MAE: Selective correspondence enhancement with masked autoencoder for self-supervised landmark estimation. In: Proc. IEEE/CVF CVPR (2024)

[6] Wang, G., et al.: FaceMAE: Privacy-preserving face recognition via masked autoencoders. arXiv:2205.11090 (2022)

[7] Gu, P., Li, H., Zhang, Y., Wang, C., Chen, D.Z.: Self pre-training with topology- and spatiality-aware masked autoencoders for 3D medical image segmentation. In: Proc. IEEE Int. Conf. Bioinformatics Biomedicine (BIBM), pp. 3608–3613 (2025)

[8] Sun, L., Lian, Z., Liu, B., Tao, J.: MAE-DFER: Efficient masked autoencoder for self-supervised dynamic facial expression recognition. In: Proc. 31st ACM Int. Conf. Multimedia, pp. 6110–6121 (2023)

[9] Guo, H., Hosseini, R., Zhang, R., Somayajula, S.A., Chowdhury, R.R., Gupta, R.K., Xie, P.: Downstream task guided masking learning in masked autoencoders using multi-level optimization. Trans. Mach. Learn. Res. (2025). https://openreview.net/forum?id=oVZ9XaOSFK

[10] Williford, J.R., May, B.B., Byrne, J.: Explainable face recognition. In: ECCV (2020)

[11] Lu, Y., Xu, Z., Ebrahimi, T.: Towards visual saliency explanations of face verification. In: IEEE WACV, pp. 4726–4735 (2024)

[12] Bulat, A., Tzimiropoulos, G.: How far are we from solving the 2D & 3D face alignment problem? In: Proc. IEEE/CVF ICCV, pp. 1021–1030 (2017)

[13] Liu, Z., Luo, P., Wang, X., Tang, X.: Deep learning face attributes in the wild. In: Proc. IEEE/CVF ICCV, pp. 3730–3738 (2015)

[14] Huang, G.B., Ramesh, M., Berg, T., Learned-Miller, E.: Labeled Faces in the Wild: A Database for Studying Face Recognition in Unconstrained Environments. In: Workshop on Faces in Real-Life Images, ECCV (2008)

[15] Cao, Q., et al.: VGGFace2: A dataset for recognising faces across pose and age. In: IEEE FG (2018)

[16] Fink, M., Perona, P.: Caltech 10k Web Faces (v1.0). CaltechDATA (2022). https://doi.org/10.22002/D1.20132

[17] Karras, T., Laine, S., Aila, T.: A style-based generator architecture for generative adversarial networks. In: Proc. IEEE/CVF CVPR, pp. 4401–4410 (2019)

[18] Karras, T., Aila, T., Laine, S., Lehtinen, J.: Progressive growing of GANs for improved quality, stability, and variation. In: ICLR (2018)